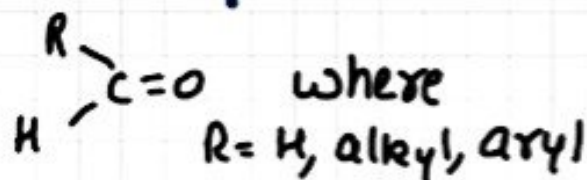
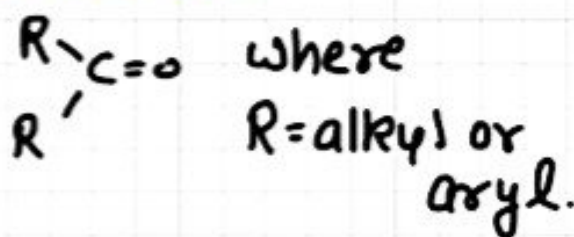


General Formula :- $C_nH_{2n}O$ having $>C=O$ gp.

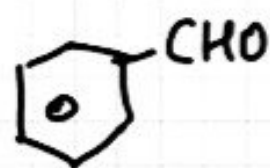
Aldehyde:



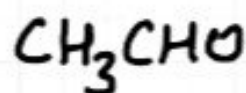
ketone:



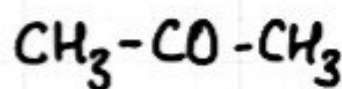
COMMON Name -



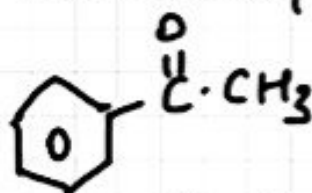
Benzaldehyde



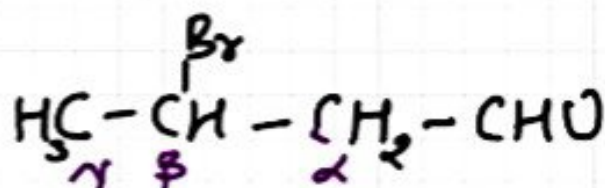
Acetaldehyde



Acetone



Acetophenone



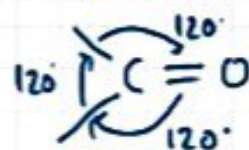
β -bromobutyraldehyde

EXAMPLE FROM NCERT:

STRUCTURE OF Carbonyl gp

Structure	Common name	IUPAC name
Aldehydes		
$HCHO$	Formaldehyde	Methanal
CH_3CHO	Acetaldehyde	Ethanal
$(CH_3)_2CHCHO$	Isobutyraldehyde	2-Methylpropanal
	7-oxoheptanal	3-Methylcyclohexanecarbaldehyde
$CH_3CH_2CH_2CHO$	α -Methylpropionaldehyde	2-Methylpropanal
$CH_3CH_2CH_2CH_2CHO$	Valeraldehyde	Pentanal
$CH_3CH=CHCHO$	Acrolein	Prop-2-enal
	Phthalaldehyde	Benzene-1,2-dicarbaldehyde
	m-Bromobenzaldehyde	3-Bromobenzaldehyde or 3-Bromobenzaldehyde
Ketones		
$CH_3COCH_2CH_2CH_3$	Methyl n-propyl ketone	Pentan-2-one
$(CH_3)_2CHCOCH_2CH_3$	Diisopropyl ketone	2,4-Dimethylpentan-3-one
	α -Methylcyclohexanone	3-Methylcyclohexanone
$(CH_3)_2C=O$	Mesityl oxide	4-Methylpent-3-en-2-one

Hybridisation and geometry

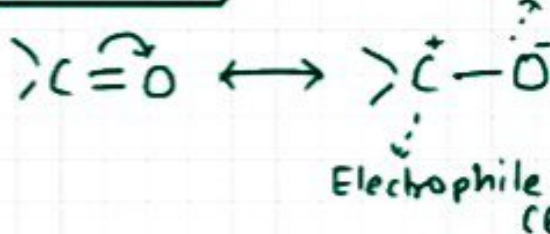


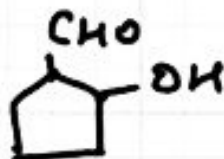
• Carbonyl carbon is sp^2 hybridized
• Trigonal planar in shape

• It contains 3 σ and 1 π bond.

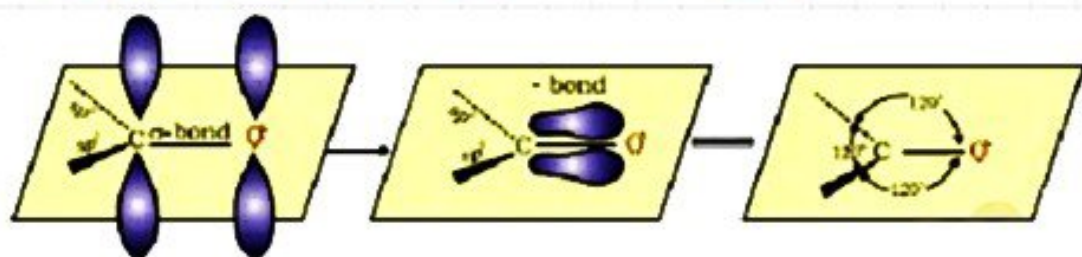
Polarity $\rightarrow \overset{+}{C}=\overset{-}{O}$ Highly polar in Nature

Resonance



- (i) α -methoxypropionaldehyde $\Rightarrow$ $\text{CH}_3 - \overset{\text{OCH}_3}{\underset{\text{O}}{\text{C}}} - \text{CH}_2 - \text{CHO}$
- (ii) 3-hydroxybutanal $\Rightarrow$ $\text{CH}_3 - \overset{\text{OH}}{\text{CH}} - \text{CH}_2 - \text{CHO}$
- (iii) 2-hydroxy cyclopentane carbaldehyde $\Rightarrow$ 

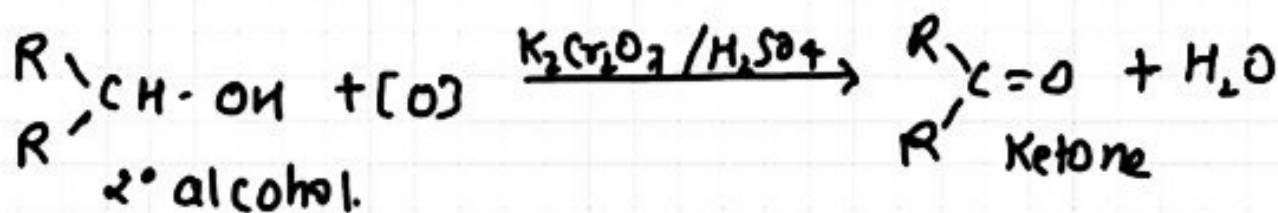
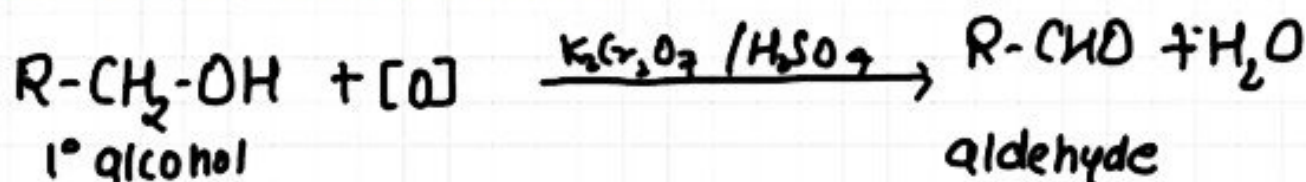
STRUCTURE OF CARBONYL GROUP



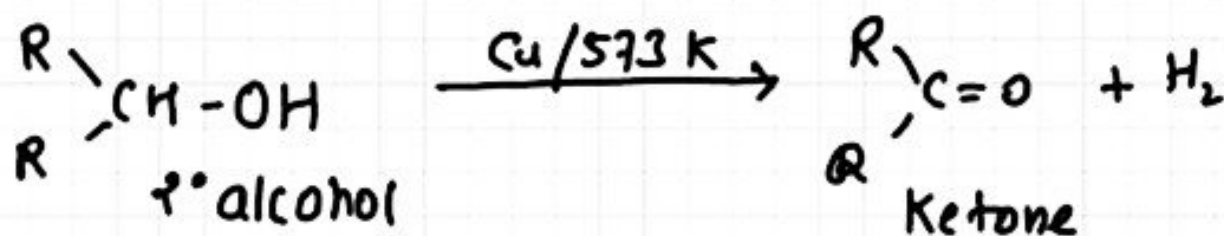
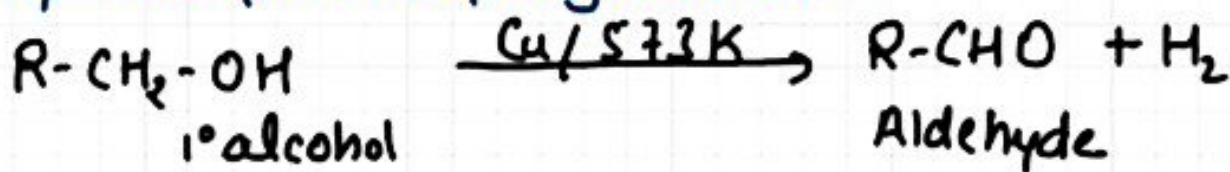
PREPARATION OF ALDEHYDE & KETONE $\Rightarrow$

FROM ALCOHOL

$\Rightarrow$ By Oxidation of Alcohol

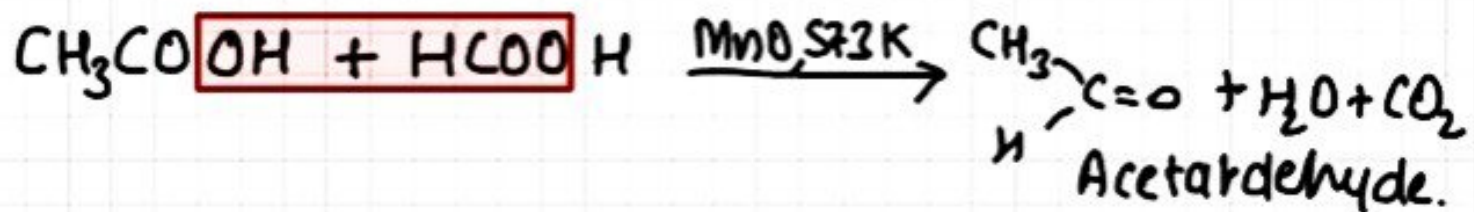
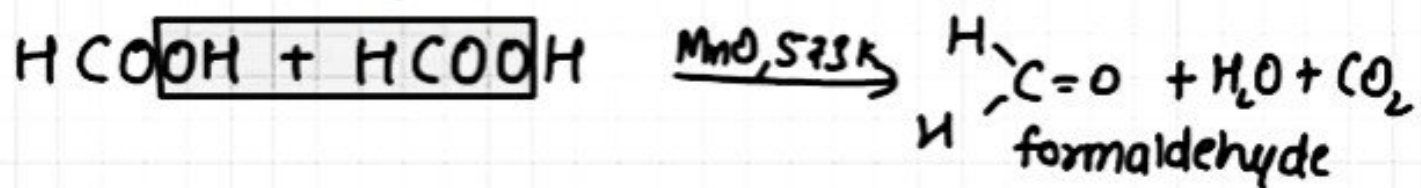


$\Rightarrow$ By Catalytic Dehydrogenation

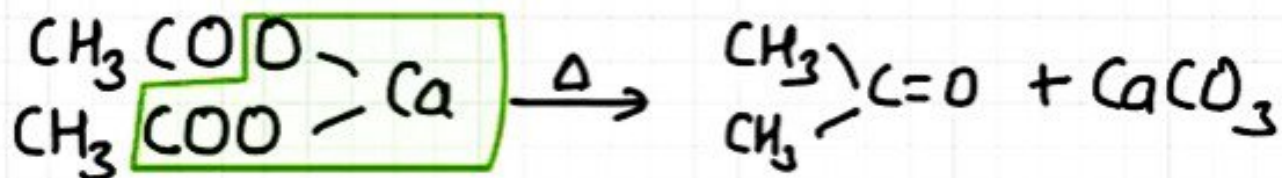
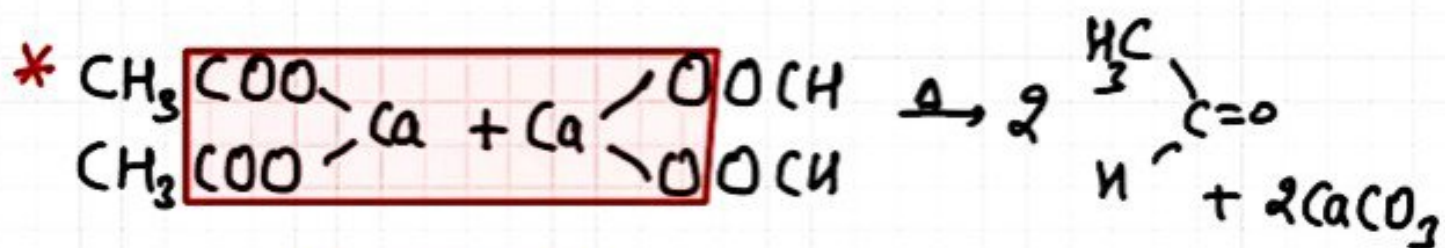
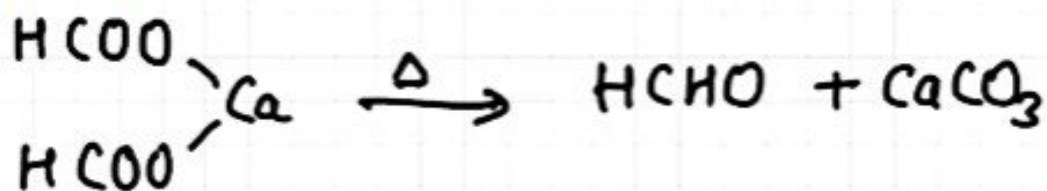


► FROM CARBOXYLIC ACID:

a) By passing vapours of MnO

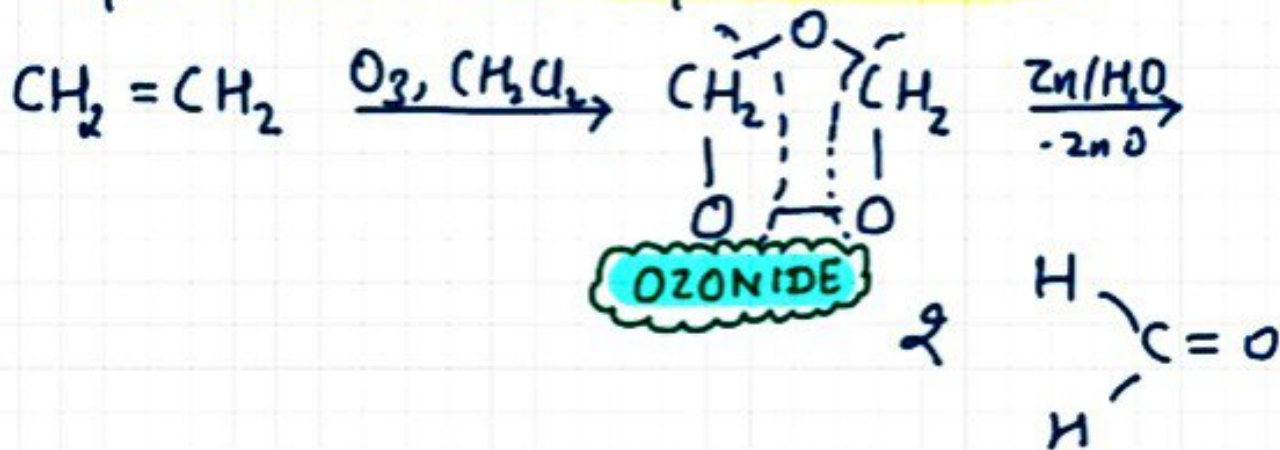


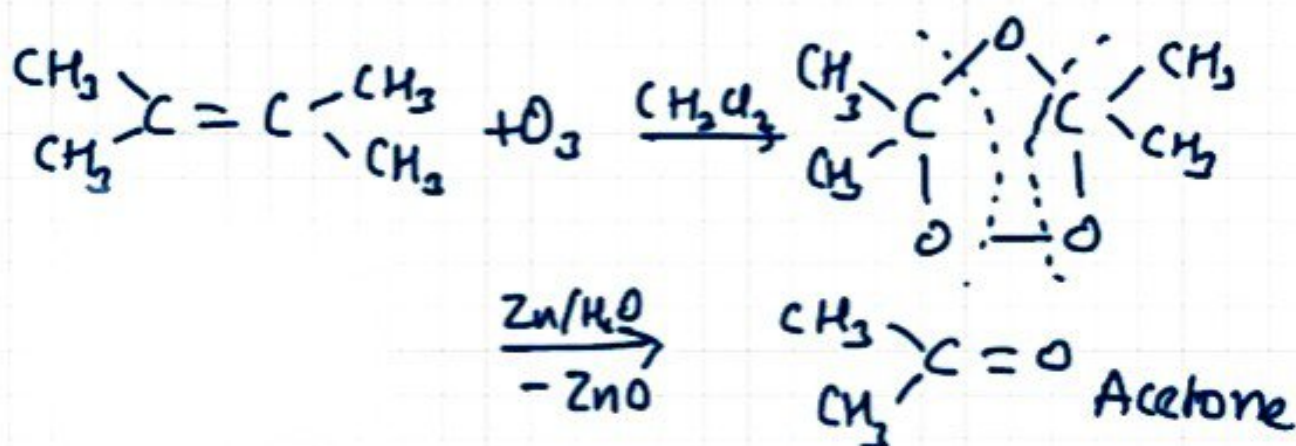
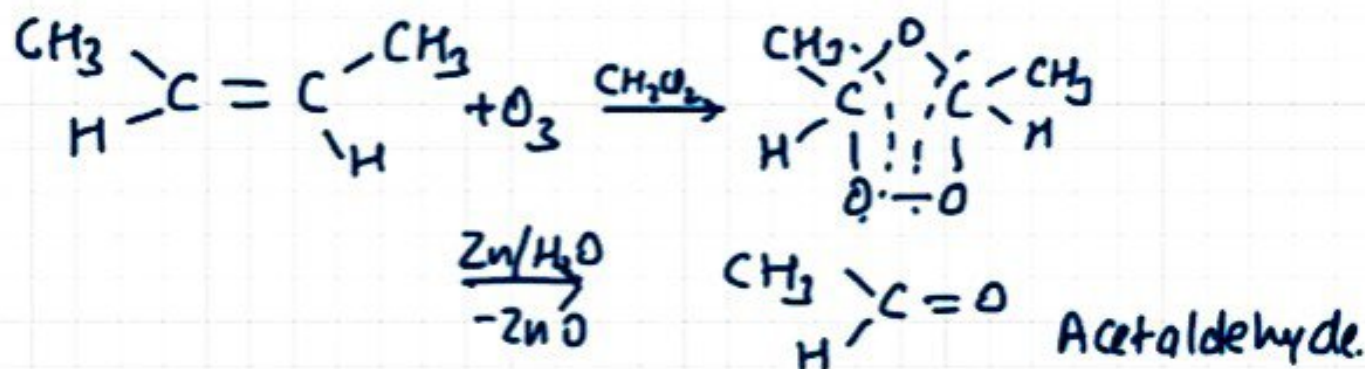
b) By dry distillation of calcium salt of fatty acid



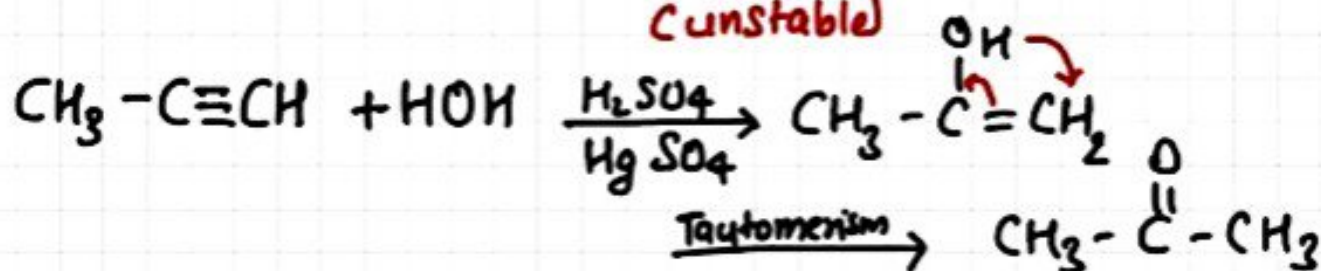
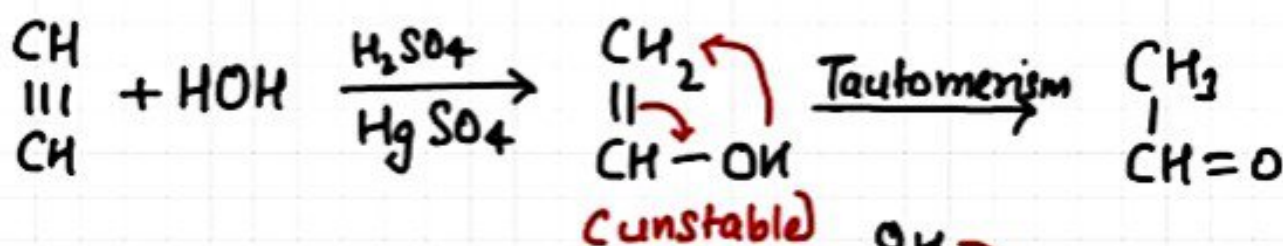
► FROM HYDROCARBONS

• By Reductive Ozonolysis of Alkenes

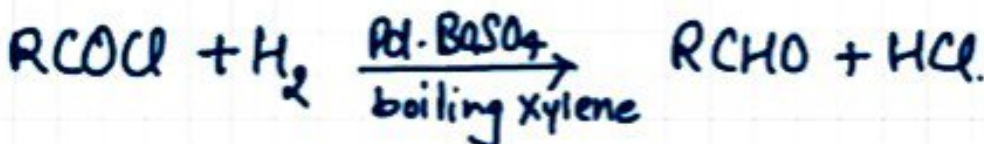
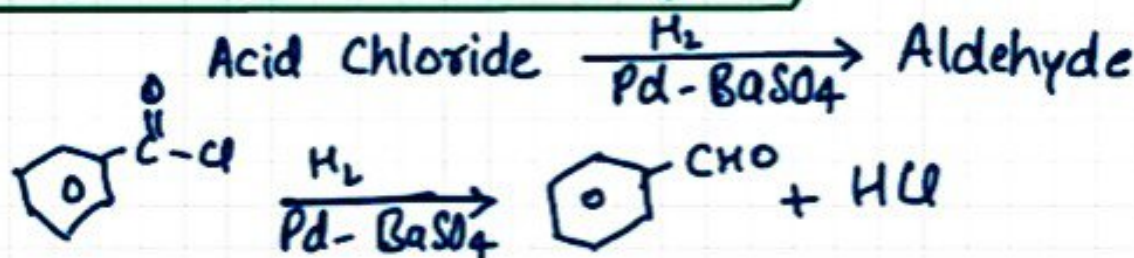




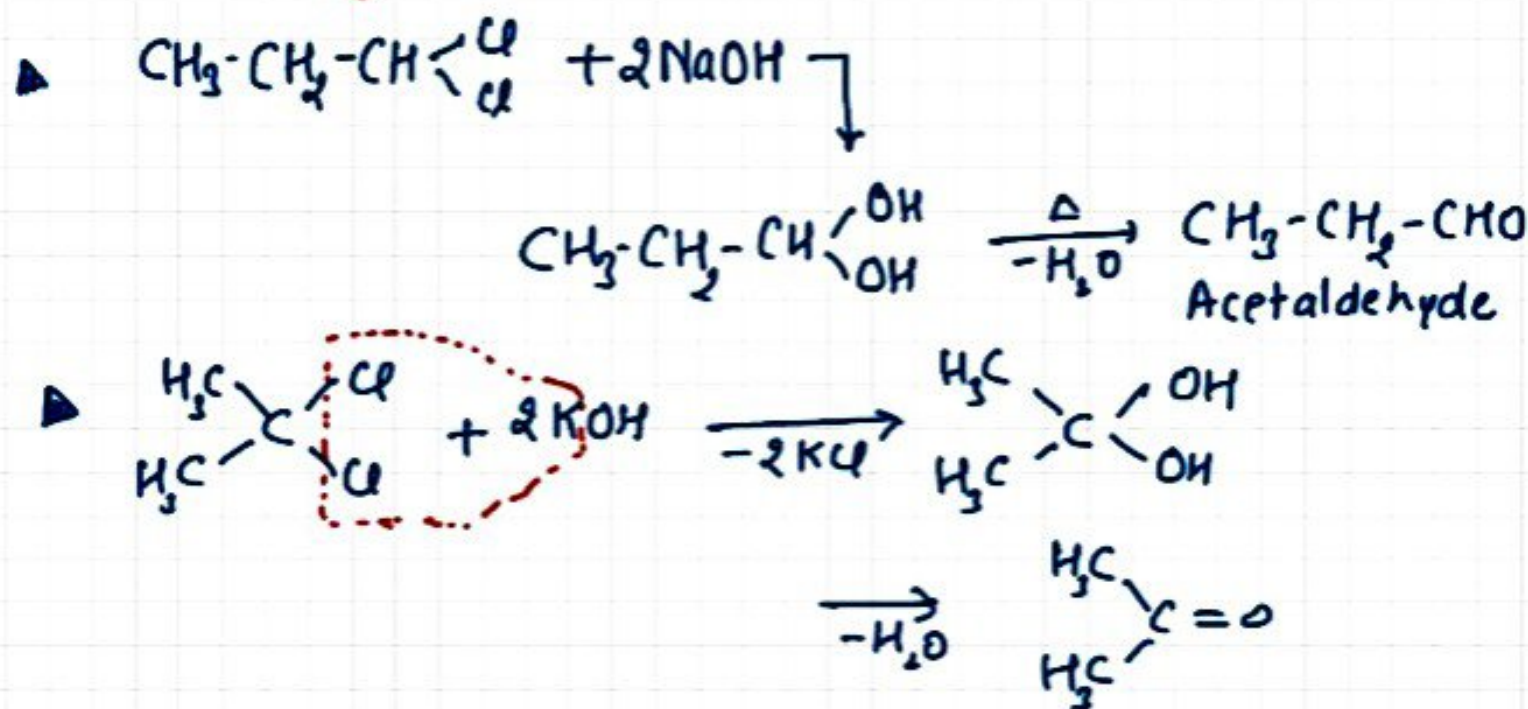
By Hydration of alkynes



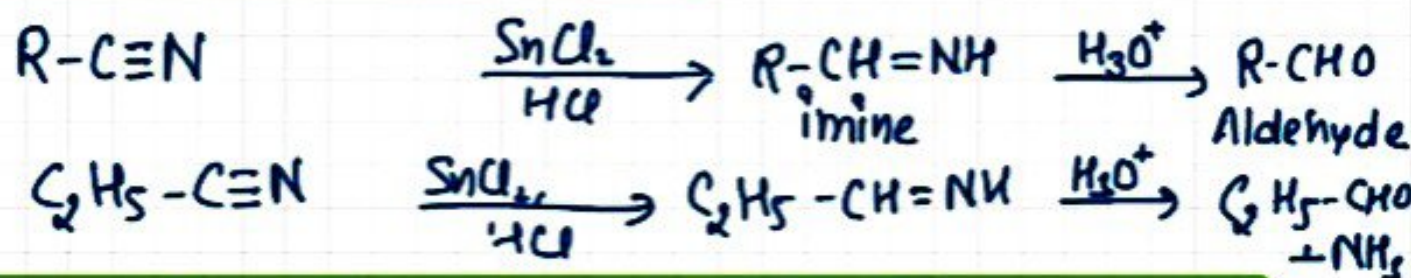
ROSENMUND REDUCTION



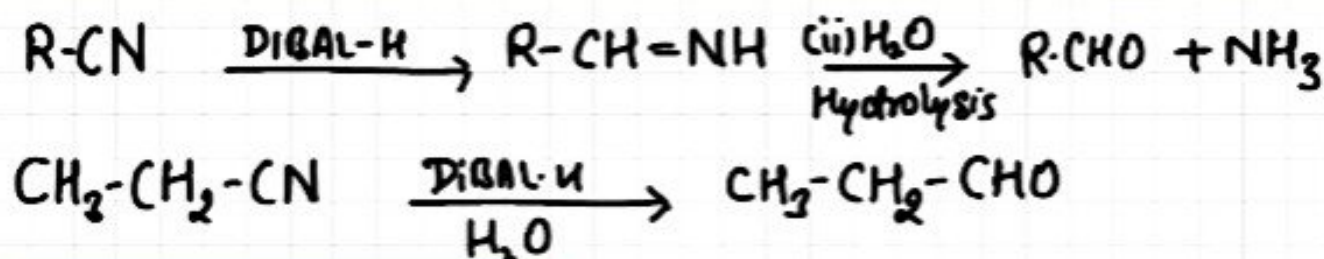
FROM gem-Dihalides



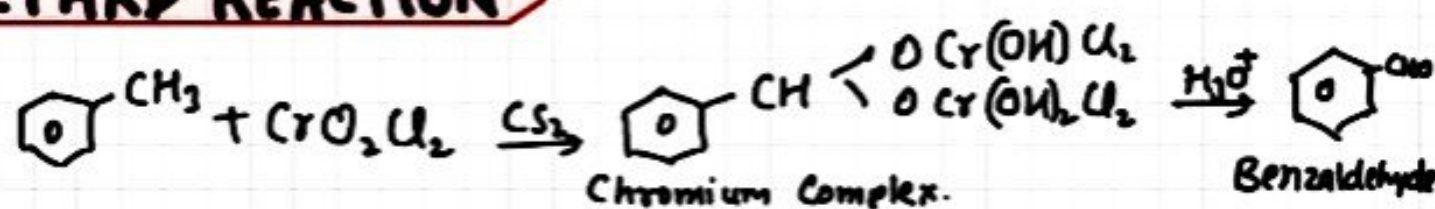
STEPHEN REDUCTION



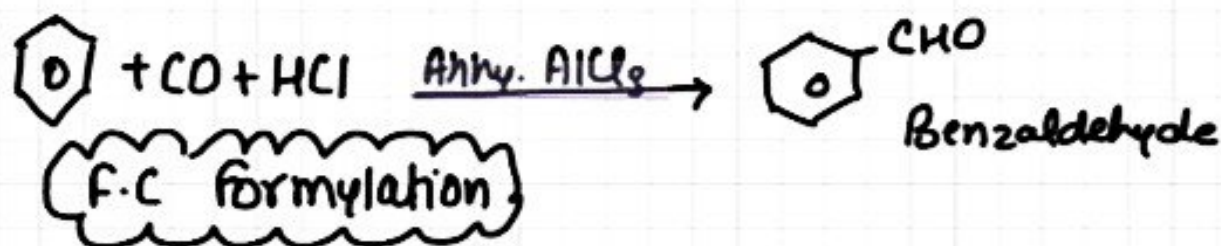
✱ DIISOBUTYL ALUMINIUM HYDRIDE
 or $[(\text{CH}_3)_2\text{CHCH}_2]_2\text{AlH}$ or $\text{AlH}(\text{i-Bu})_2$
 or DIBAL-H



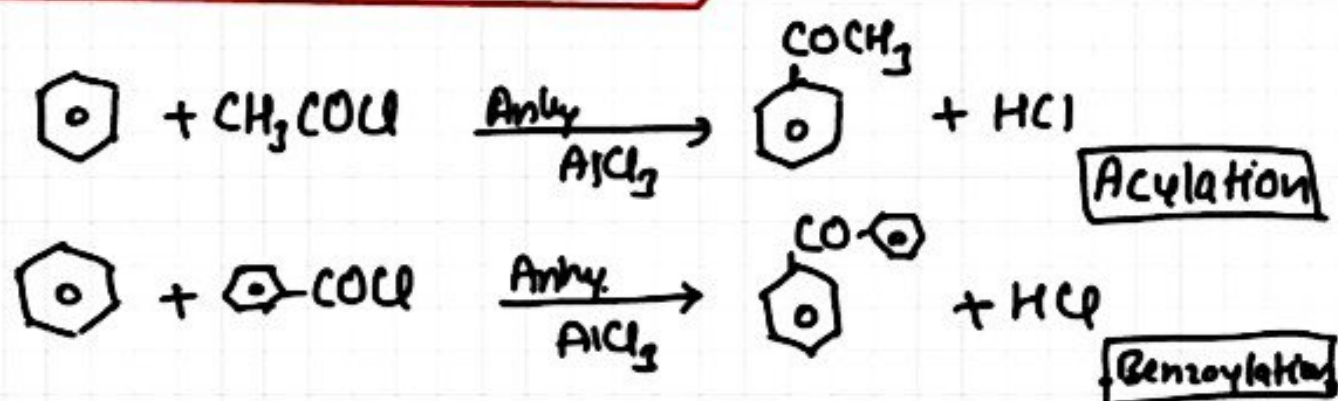
ETARD REACTION



GATTERMANN KOCH REACTION



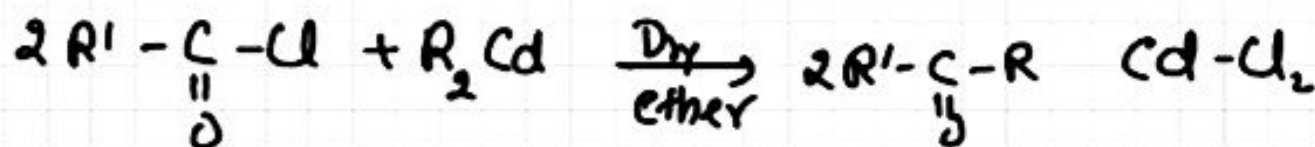
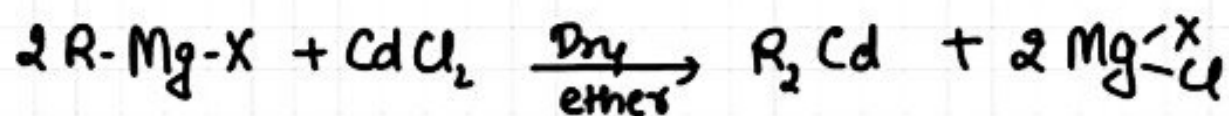
FRIEDEL CRAFT REACTION



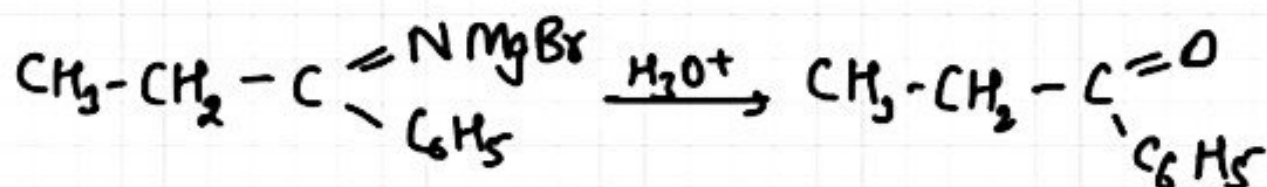
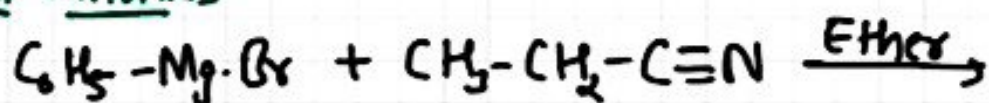
PREPARATION OF KETONES

From Acid Chlorides

Acyl chloride, on treatment with dialkyl cadmium gives Ketones



From Nitriles



PHYSICAL PROPERTIES

✓ PHYSICAL STATE

At room temp, methanal is a gas, ethanal is a volatile liquid. Other aldehyde and ketones are liquid or solid.

✓ ODOUR

- The lower aldehydes have sharp pungent odour.
- The odour becomes less pungent and more fragrant as the size of molecules increases. These are used in perfumes and flavouring agents.

✓ BOILING POINT

The b.pt of aldehydes and ketones are higher than that of hydrocarbons and ethers due to stronger dipole-dipole interactions in them but lower than alcohols due to the absence of inter-molecular H-bonding.

Ketones are more polar than aldehyde. Therefore the b.pt of ketones are higher than that of aldehyde.

✓ SOLUBILITY

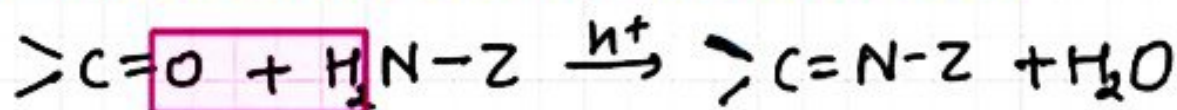
• Lower aldehydes and ketones are miscible with water in all proportions due to their ability to form H-bond with water.

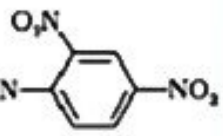
solubility decreases with increase in size of alkyl group.

Aldehyde and ketones are soluble in organic solvents like benzene, ether, methanol, chloroform.

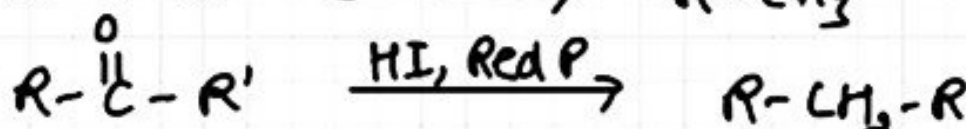
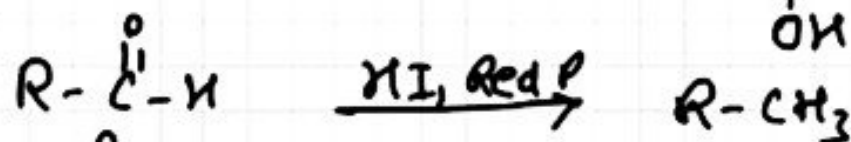
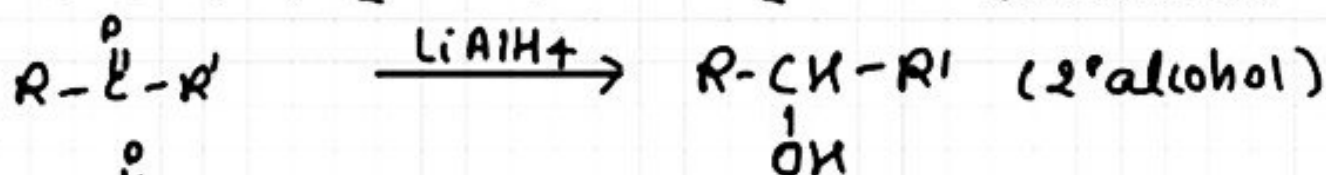
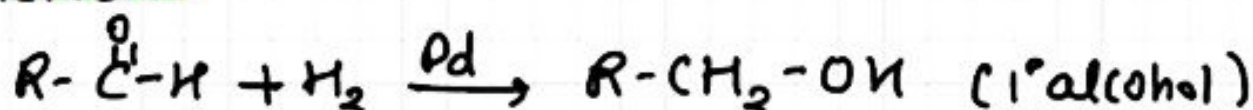
CHEMICAL PROPERTIES

✓ ADDITION OF AMMONIA & ITS DERIVATIVES

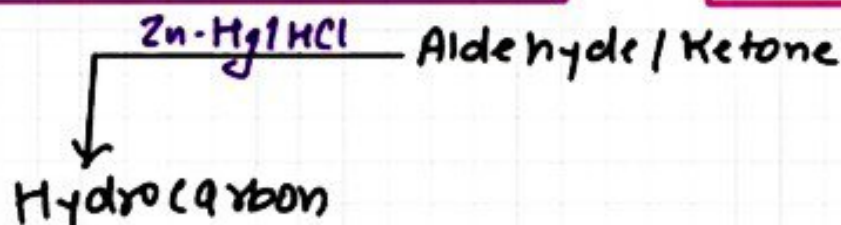


Z	Reagent name	Carbonyl derivative	Product name
-H	Ammonia	$>C=NH$	Imines
-R	Amine	$>C=NR$	Substituted imine (Schiff's base)
-OH	Hydroxylamine	$>C=N-OH$	Oxime
-NH ₂	Hydrazine	$>C=N-NH_2$	Hydrazone
-HN- 	Phenylhydrazine	$>C=N-NH-$	Phenylhydrazone
-HN- 	2,4-Dinitrophenylhydrazine	$>C=N-NH-$	2,4 Dinitrophenylhydrazone
-NH-C(=O)-NH ₂	Semicarbazide	$>C=N-NH-C(=O)-NH_2$	Semicarbazone

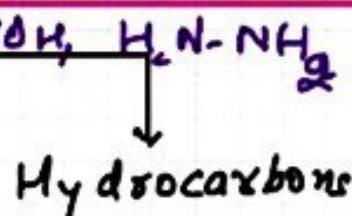
✓ Reduction



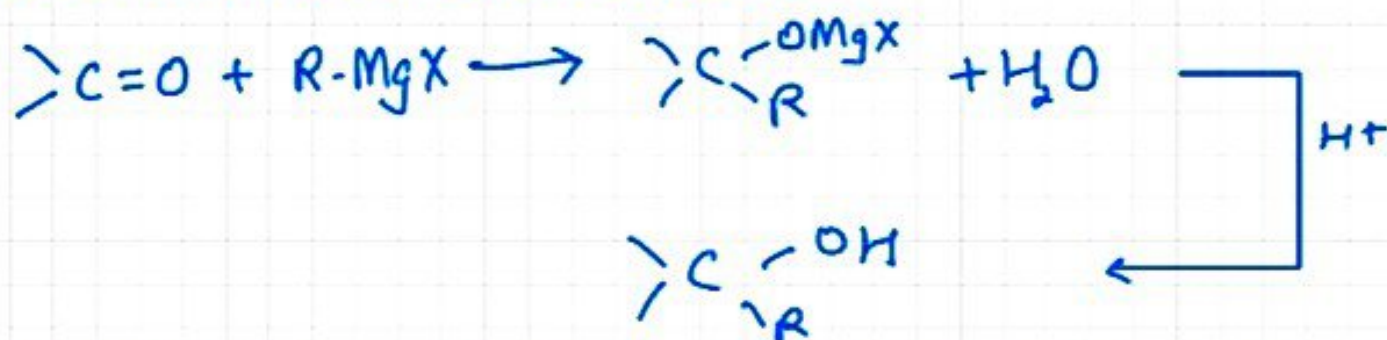
CLEMMENSEN REDⁿ



WOLFF-KISHNER REDⁿ

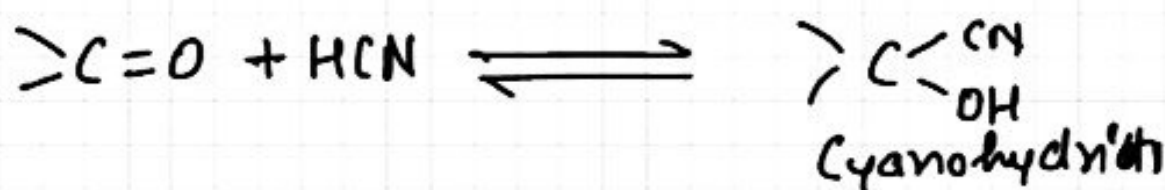


✓ ADDITION OF GRIGNARD REAGENT

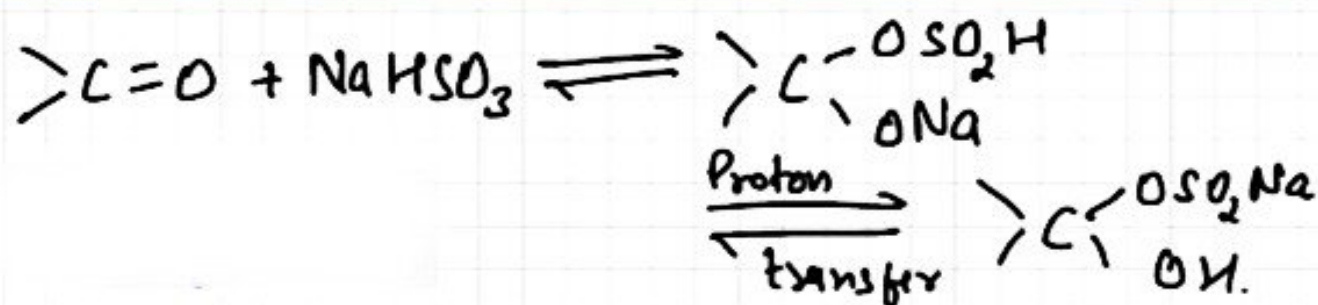


One of the best method to prepare 1°, 2°, 3° alcohol.

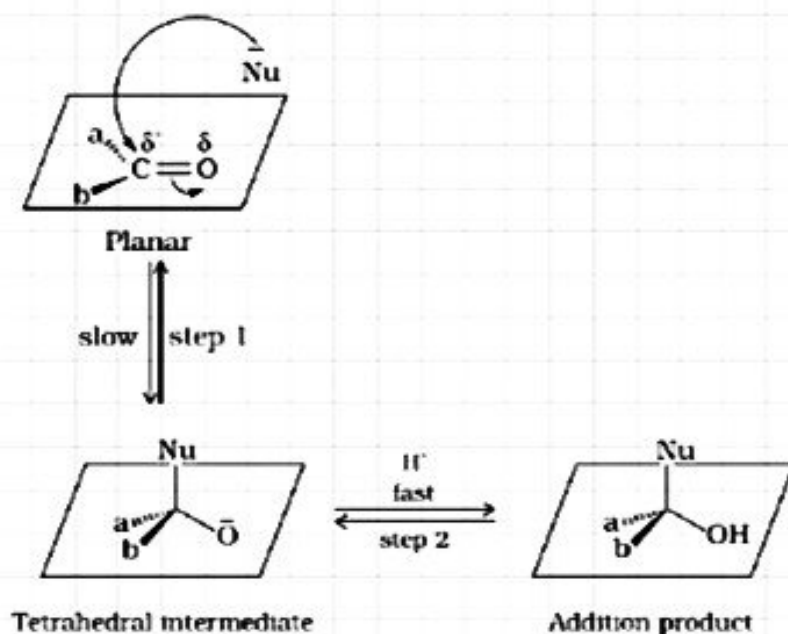
✓ ADDITION OF HYDROGEN CYANIDE (HCN)



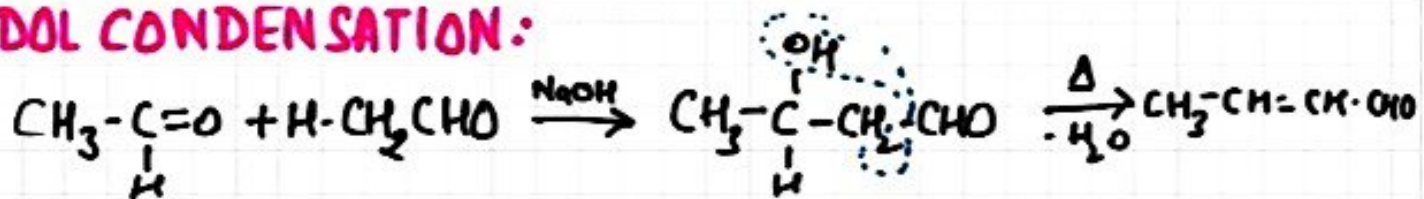
✓ ADDITION OF SODIUM HYDROGEN SULPHITE



✓ MECH. OF NUCLEOPHILIC ADD. REACTION



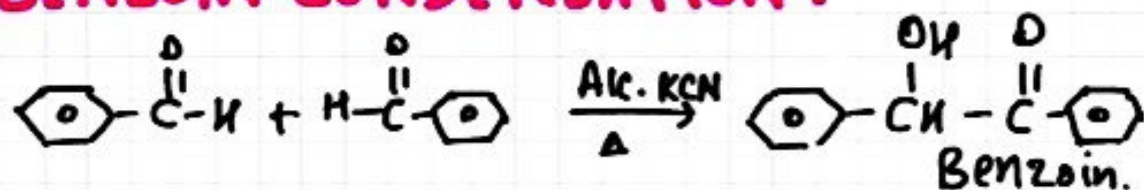
a) ALDOL CONDENSATION:



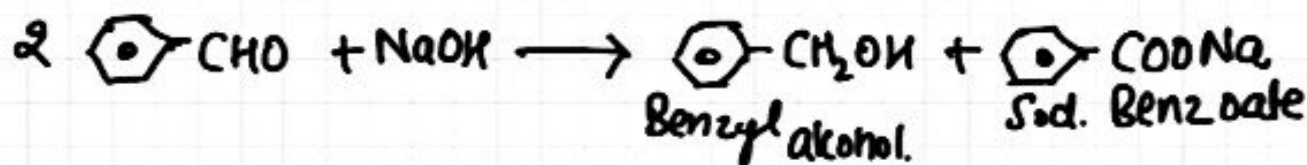
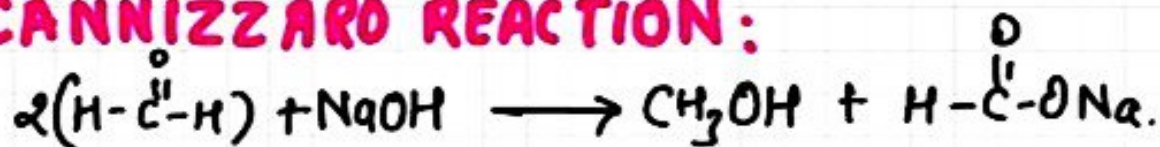
b) CROSS ALDOL CONDENSATION:



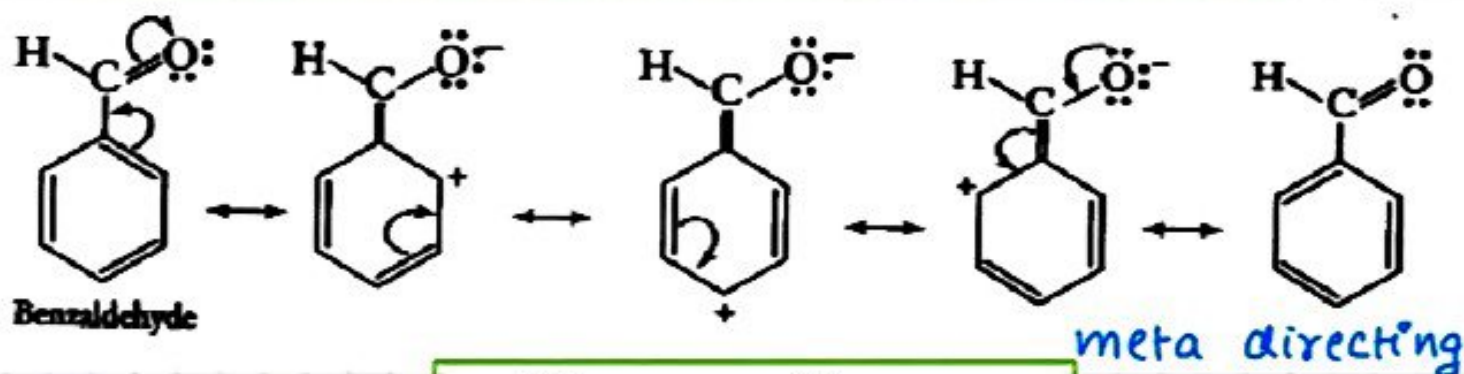
c) BENZOIN CONDENSATION:



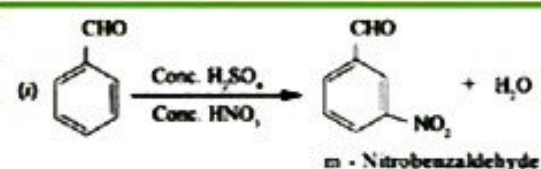
d) CANNIZZARO REACTION:



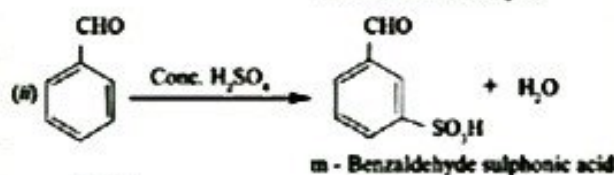
► Resonance in Benzaldehyde / Acetophenone



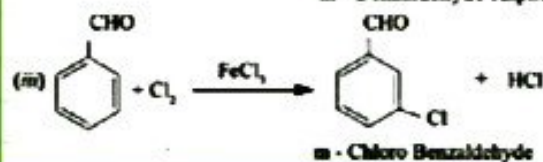
NITRATION



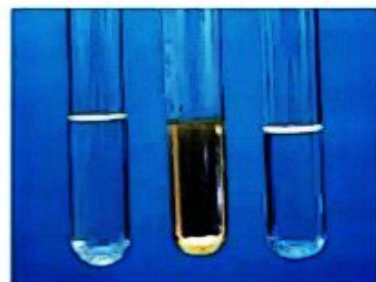
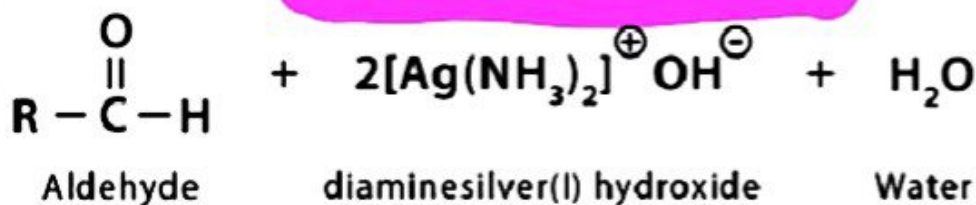
SULPHONATION



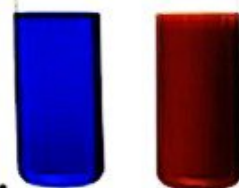
CHLORINATION



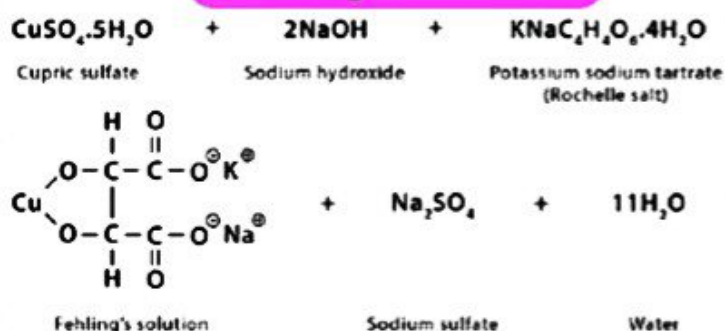
Tollens' Test



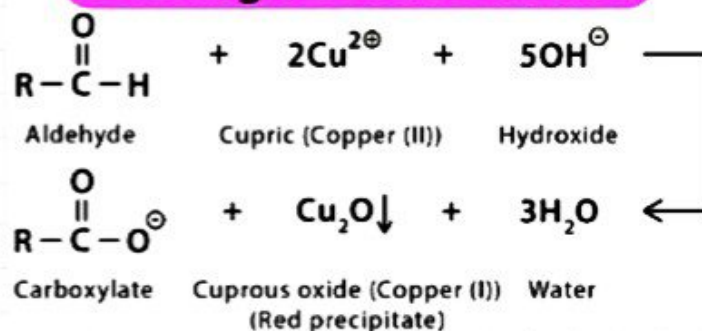
F. Test



Fehling's Solution



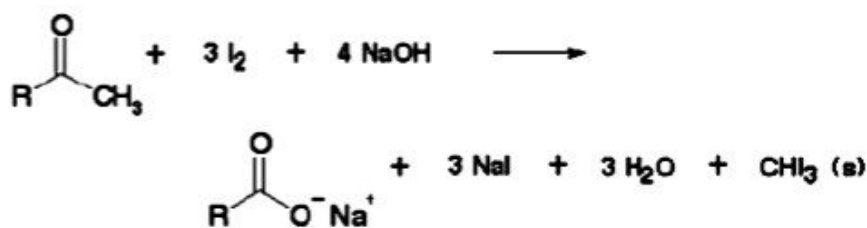
Fehling's Test Reaction



IODOFORM TEST

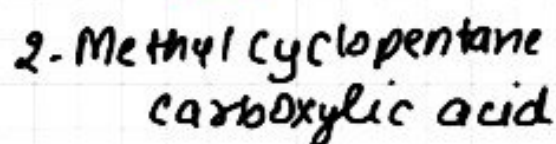
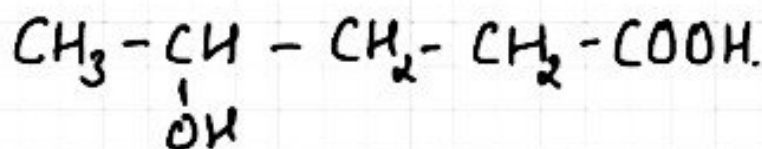
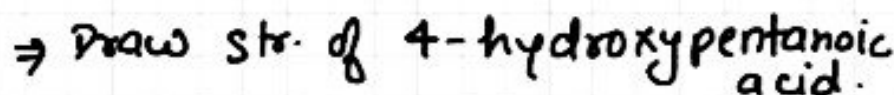
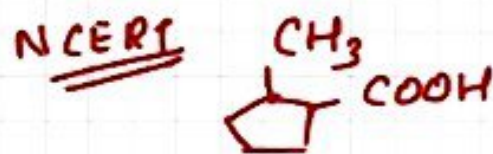
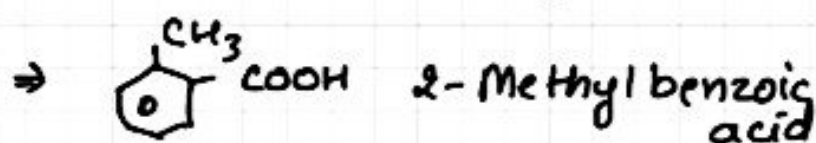
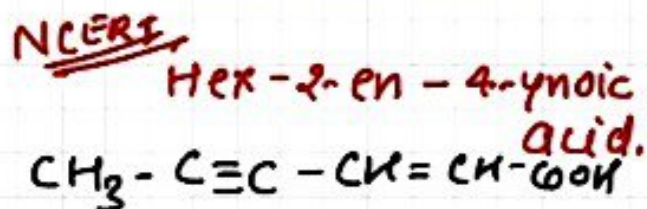
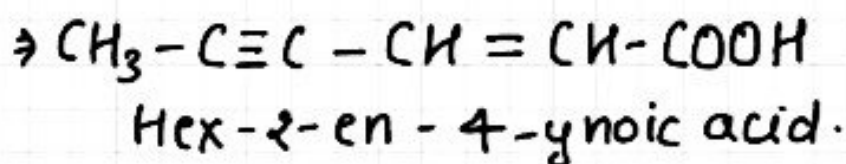
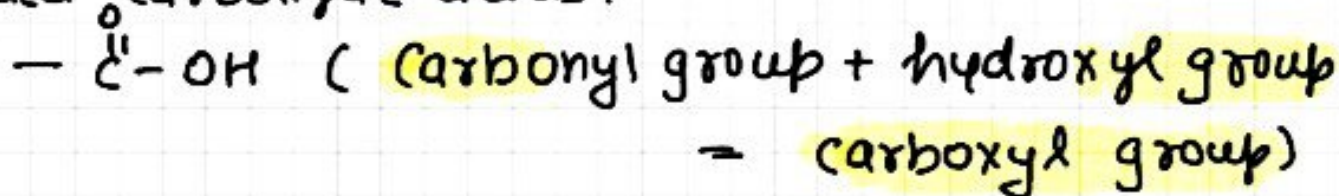


Give positive result for ethanal and $-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ (Methyl Ketone)

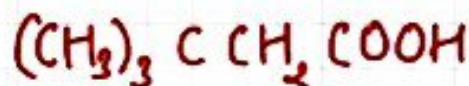


CARBOXYLIC ACIDS

Carbon compounds containing carboxyl group ($-\text{COOH}$) are called carboxylic acids.



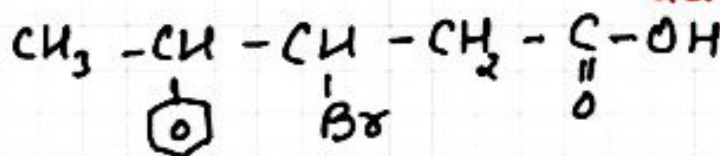
NCERT Give IUPAC Name of



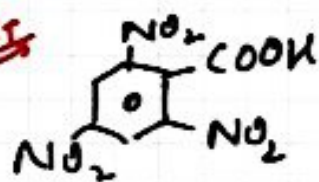
3,3-Dimethylbutanoic acid.

NCERT

ii) 3-Bromo-4-phenylpentanoic acid.



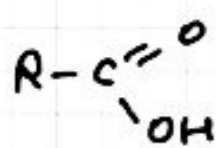
NCERT



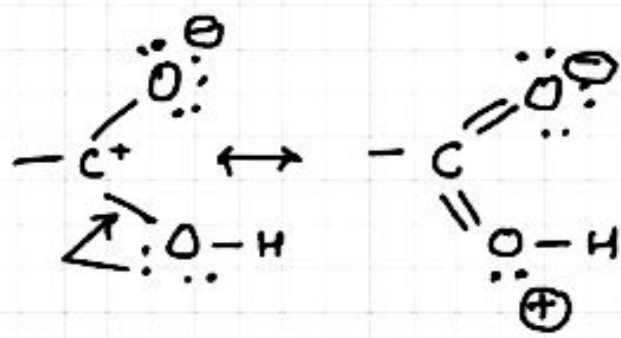
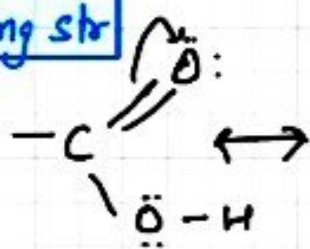
2,4,6-Trinitro benzoic acid.

Structure of Carboxyl group.

Resonating str



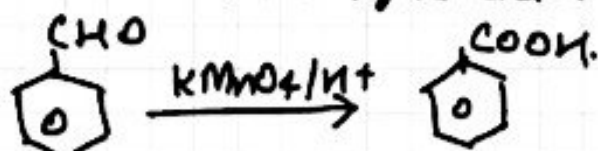
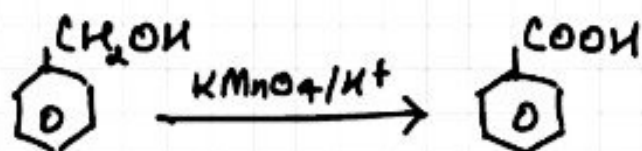
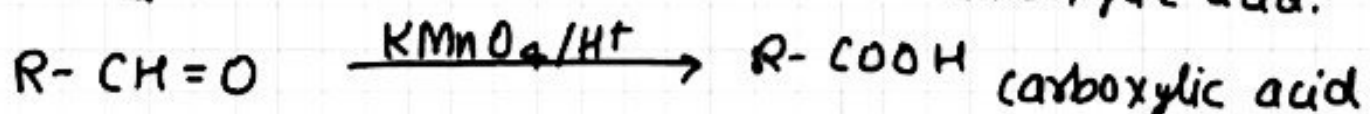
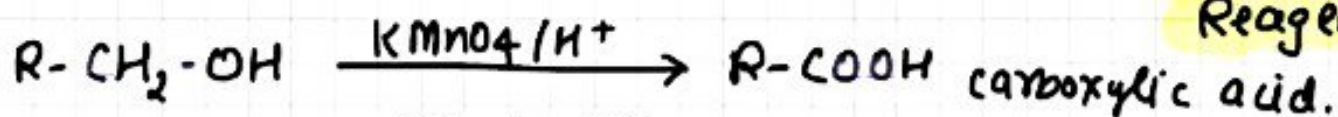
- ✓) sp^2 hybridised
- ✓) Trigonal planar
- ✓) 120° bond angle



METHODS OF PREPARATION OF CARBOXYLIC ACIDS

From 1° Alcohols & Aldehyde

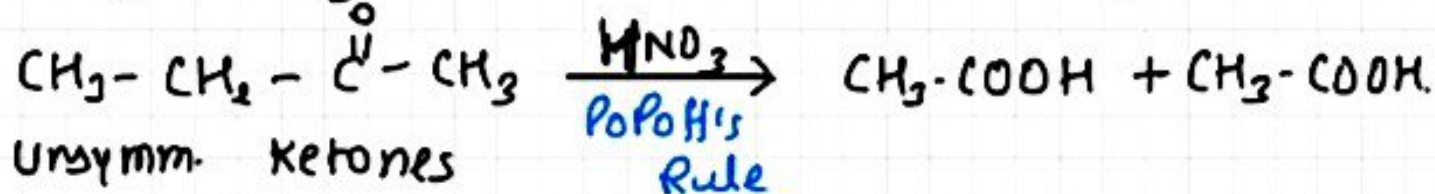
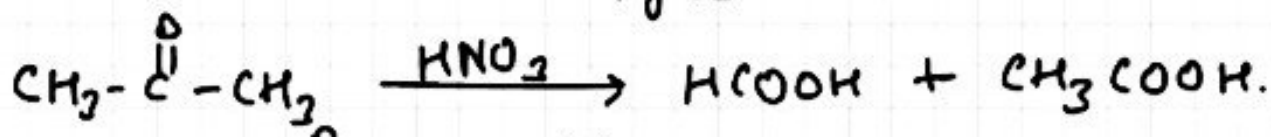
- KMnO_4 in neutral, acidic or alkaline medium
- $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium
- Chromium trioxide (CrO_3) in acid. med (Jones Reagent)



- Aldehyde can also be oxidised by mild oxidising agent like Fehling solution or Tollens Reagent.

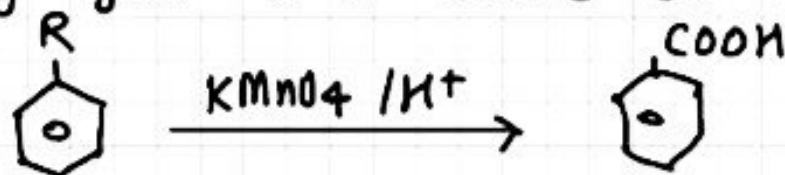
From Ketones

Under drastic conditions, ketones can be converted into carboxylic acid.



From Alkyl benzene

C-chain on benzene ring is oxidised to $-\text{COOH}$ group by using strong oxidising agent like acidic or alkaline KMnO_4



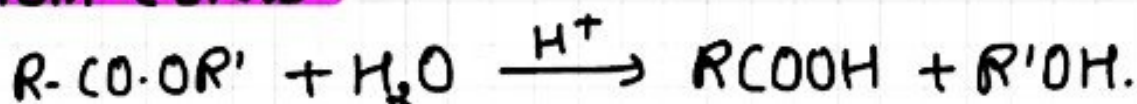
FROM ACID DERIVATIVES

acid derivative $\xrightarrow{\text{Hydrolysis}}$ carboxylic acid.

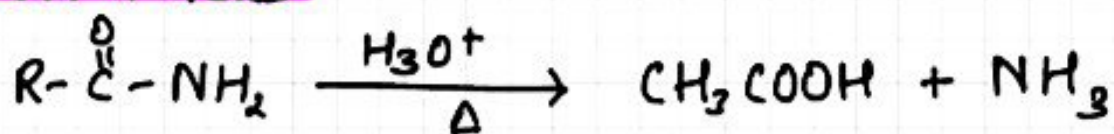
✓ From Acid Halide



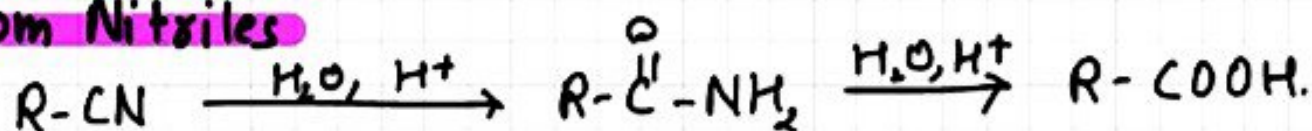
✓ From esters



✓ From Amides



✓ From Nitriles



FROM GRIGNARD REAGENTS

Carboxylic acids are prepared by the action of Grignard Reagent on CO_2 followed by hydrolysis.



PHYSICAL PROPERTIES OF CARB. ACIDS

✓ PHYSICAL STATE

• Aliphatic carboxylic acids upto 9-C atoms are colourless liquids.

• The higher Acids are wax like solids.

✓ ODOUR:

• The lower carboxylic acids have unpleasant odour.

• Higher Carboxylic Acids are odourless.

✓ BOILING POINT

• For the compounds of comparable mol. mass, the increasing order of B.Pt is...

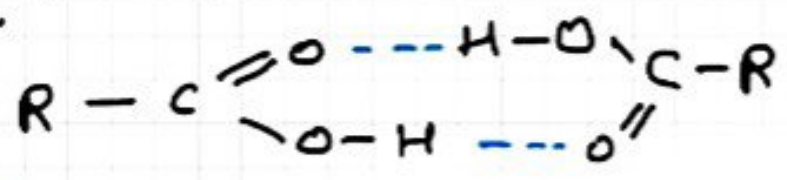
Hydrocarbons < Haloalkanes < Ethers < Aldehyde < Ketones < Amines < Alcohol < Carboxylic Acid.

Reason of higher B.Pt of Carboxylic Acid.

- (i) carboxylic acid molecules are more extensive associated through intermolecular Hydrogen bonding.
- (ii) The electron withdrawing nature of carbonyl group increases the polarity of O-H bond, which makes the H-bonding stronger than alcohol.

DIMER FORMATION

Most of the carboxylic acids exist as dimer in the vapour phase or in the aprotic solvents.

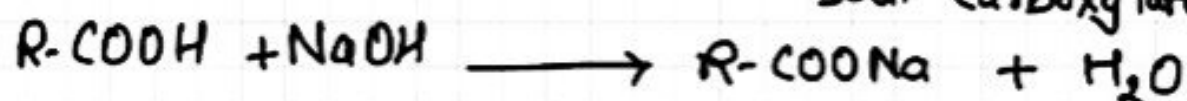


✓ SOLUBILITY

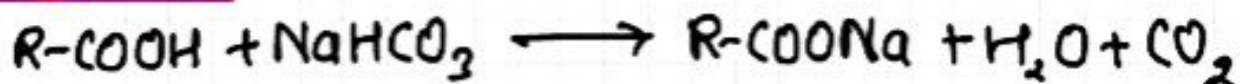
- Aliphatic carboxylic acids having upto 4-C atoms are miscible in water in all proportions due to the formation of H-Bond with water.
- The solubility decreases with increasing no. of carbon atoms.
- Aromatic carboxylic acids like benzoic acid is nearly insoluble in cold water.
- As Hydrocarbon increases, solubility of carboxylic acid decreases in water.

✓ CHEMICAL PROPERTIES OF CARBOXYLIC ACIDS

✓ ACIDITY OF ALCOHOLS / Rxn with Metals

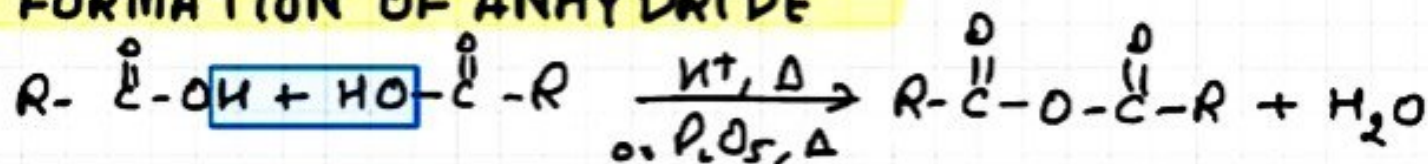


NaHCO₃ Test

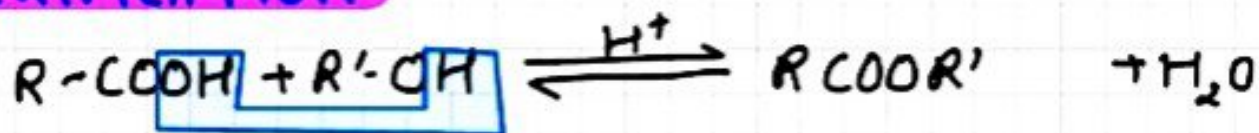


Sod. Hydrogen carbonate (NaHCO₃) is used as a characteristic test for carboxylic acids. All carboxylic acids give brisk effervescence of CO₂ gas.

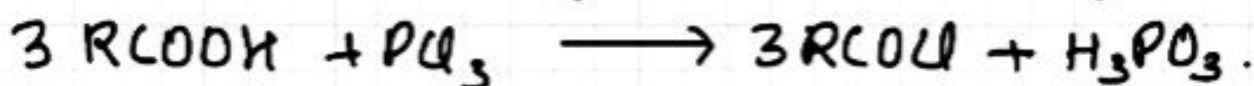
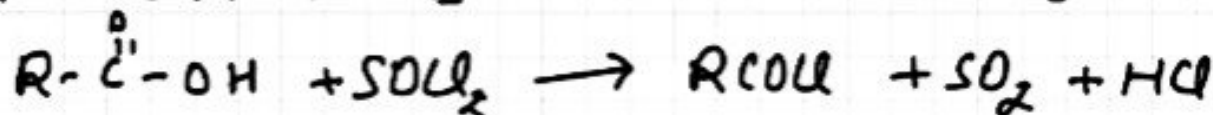
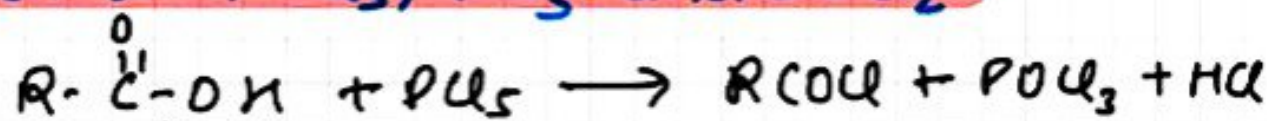
✓ FORMATION OF ANHYDRIDE



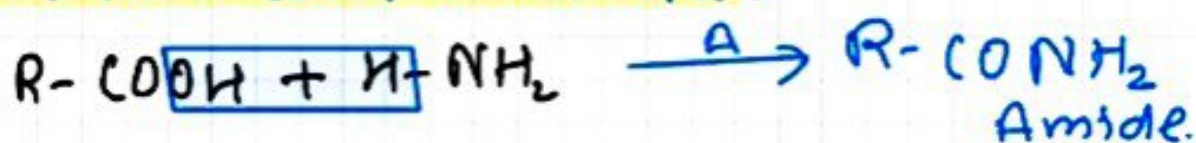
✓ ESTERIFICATION



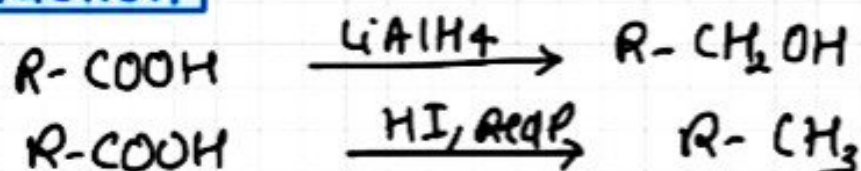
✓ Reaction with PCl₅, PCl₃ and SOCl₂:



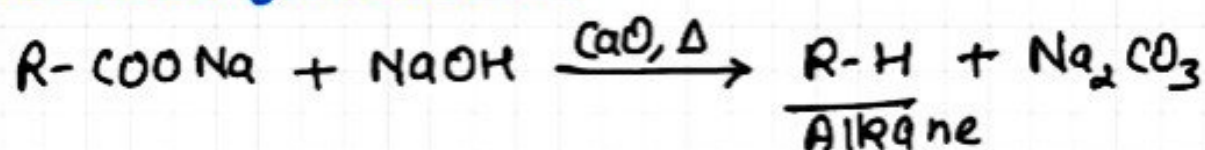
✓ Reaction with Ammonia:



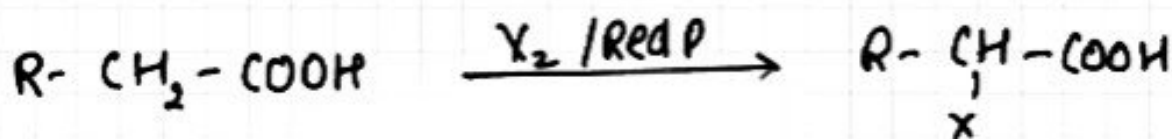
✓ Reduction



✓ Decarboxylation Rxn

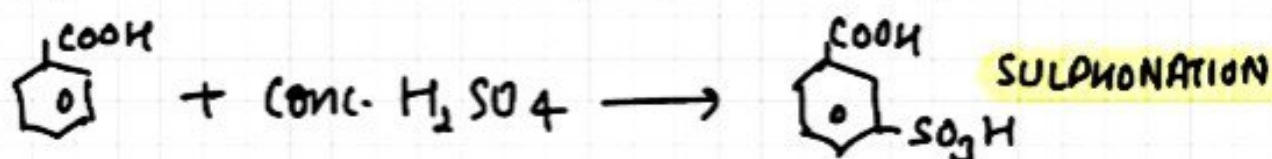
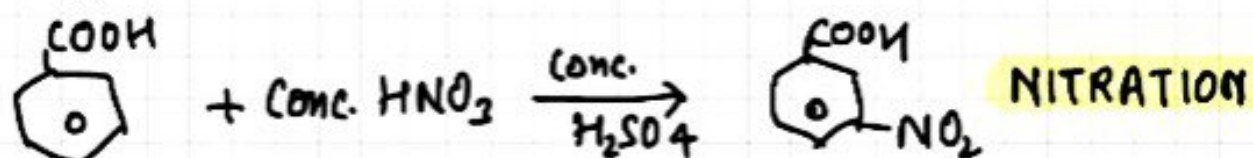
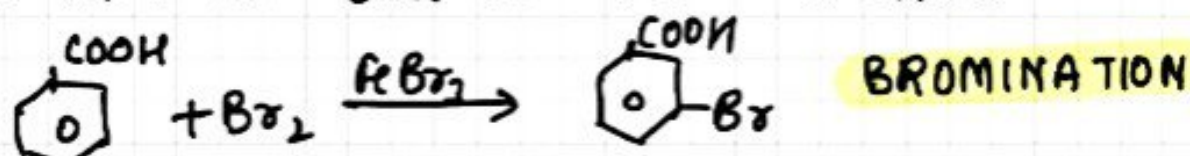


H.V.Z. REACTION < HELL VOLHARD ZELINSKY RXN >



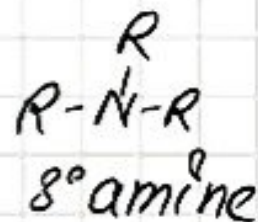
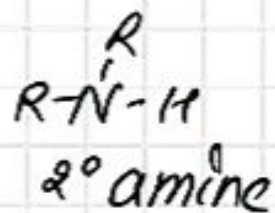
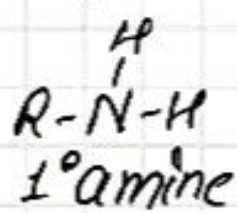
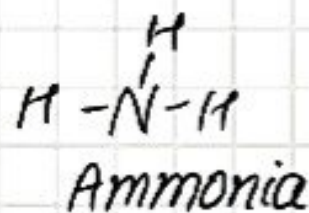
▶ ELECTROPHILIC SUBSTITUTION REACTION

The carboxyl group on benzene ring acts as a deactivating and meta directing group for electrophilic substitution Rxn due to its $-R$ effect.



(Note) Aromatic carboxylic acids, however do not undergo Friedel craft reaction because the carboxyl group is deactivating and catalyst anhydrous $AlCl_3$ bonded to the carboxyl group

These are alkyl or aryl derivatives of Ammonia



Structure of Amines

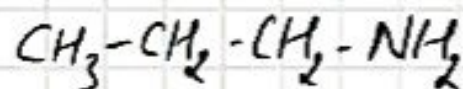


Structure - trigonal pyramidal

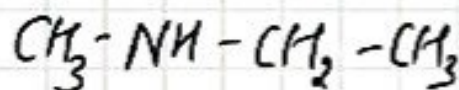
3 Bond Pair + 1 Lone Pair

Hybrid $\rightarrow sp^3$

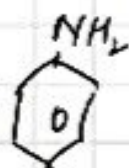
Nomenclature $\Rightarrow$



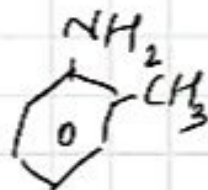
Propan-1-amine



N-Methyl ethanamine



Aniline



2-Amino toluene